Sheng'en Li

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Education

Duke Kunshan University & Duke University Expected Dec. 2026
B.S. in Applied Mathematics and Computer Science (dual-degree program) **GPA: 3.8/4.0**

- **Honors:** Dean's List with Distinction (Fall 2023, Spring 2024); Summer Research Scholar (2024, 2025).

Research Interests

Reliable machine learning; counterfactual inference and fairness in adaptive decision systems; graph and temporal learning; verification of learning-enabled systems.

Publications

[1] **Sheng'en Li**, Dongmian Zou. "COPF: An Online Framework for Deployment-Stable Counterfactual Fairness in Evolving Graphs." *International Conference on Machine Learning (ICML)*, 2026. **First author.** [paper]

[2] Shouju Wang*, Yuchen Song*, **Sheng'en Li***, Dongmian Zou. "Enhancing Fairness in Autoencoders for Node-Level Graph Anomaly Detection." *European Conference on Artificial Intelligence (ECAI)*, 2025. **Co-first author.** [paper]

Submitted Manuscript

Juntang Wang*, **Sheng'en Li***, Yihan Wang*, Jiayu Gao*, Duanduan Wang*, Dongmian Zou, Shixin Xu. "Auditing and Refining Partial-Physics Models: Scaffold Adequacy, Local Identifiability and ECG-Based Electrolyte Inference." *Submitted to Artificial Intelligence in Medicine.* **Co-first author.**

*Equal contribution.

Selected Research Experience

Counterfactual Learning for Adaptive Decision Systems **Jan. 2025 – Present**
Independent research, Duke Kunshan University / Duke University; advised by Prof. Dongmian Zou and Prof. Fan Wei

- Developed **COPF**, a decision-layer framework for fair link recommendation on evolving graphs, combining exploration and propensity logging, graph-aware doubly robust estimation, and online fairness calibration under deployment feedback.
- Extending this work in **COMET** to study counterfactual moments under weak overlap and dependent feedback.
- **Outcomes:** COPF accepted to **ICML 2026** (first author); COMET in preparation (first author).

Taylor-Model Reachability for Neural-Network-Controlled Systems **May 2026 – Present**
Research collaboration with Prof. Huan Zhang's group, University of Illinois Urbana-Champaign

- Developing a PyTorch Taylor-model engine for nonlinear ODE reachability, combining interval arithmetic, Picard iteration, and validated remainder propagation to support safety verification of neural-network-controlled systems.
- Built step-level comparisons against Flow*, a validated ODE reachability tool, to diagnose polynomial truncation and remainder growth and assess enclosure tightness, runtime, and CPU/GPU consistency.

Partial-Physics Models for ECG-Based Electrolyte Inference **Mar. 2026 – Aug. 2026**
Collaborative research, Duke Kunshan University; advised by Prof. Shixin Xu and Prof. Dongmian Zou

- Co-developed methods to assess physical-scaffold adequacy and local identifiability in ECG-based electrolyte inference, using projective alignment, rank-matched null models, and tangent/Fisher diagnostics.
- Built and validated MIMIC-IV-ECG pipelines and evaluated potassium-specific reduced-rank refinements on held-out data.
- **Outcome:** Paper submitted to *Artificial Intelligence in Medicine*.

SELECTED RESEARCH EXPERIENCE (CONTINUED)

Contextual Information and Alarm Utility in Time-Series Anomaly Detection **2025 – 2026**
Signature Work, Duke Kunshan University; advised by Prof. Dongmian Zou

- Designed matched-block experiments to isolate the value of temporal context; proved chance-level ranking for fixed current-only detectors and evaluated a covariance-based Markov model trained on normal data.
- Compared threshold and accumulation-based alarm rules on identical score streams in two simulated settings and across 119 real series from five sources, measuring event recall, detection delay, and warnings during normal periods.

Fair Graph Anomaly Detection **Jul. 2024 – May 2025**
Summer Research Scholar, Duke Kunshan University; advised by Prof. Dongmian Zou

- Developed **DECAF-GAD**, a fairness-aware graph autoencoder framework with causal disentanglement and counterfactual regularization; evaluated fairness and detection performance through multi-dataset experiments and ablations.
- **Outcome:** Co-first-author paper accepted to **ECAI 2025**.

INDUSTRY EXPERIENCE

Quantitative Research Intern **Sep. 2026 – Present**
Shangchen Private Fund, Shanghai, China

- Built liquidity-aware factor backtests with volume-participation limits, partial fills, and order carryover; measured completion rates, execution delays, and return differences across portfolio sizes.
- Developing methods for macroeconomic consensus forecasts, aligning indicator definitions and report timestamps, enforcing point-in-time availability, and assessing historical coverage across research sources.

Quantitative Developer Intern **Jul. 2026 – Sep. 2026**
Shangchen Private Fund, Shanghai, China

- Developed an internal financial LLM agent for financial-document retrieval and database-backed question answering, with traceable sources and consistent metric definitions.
- Helped build the firm's backtesting database by implementing Python/SQL data pipelines and validation checks for financial and market data.

LEADERSHIP AND SERVICE

Resident Assistant, Duke Kunshan University	Aug. 2024 – May 2025
Volunteer Mathematics Instructor, Tongren, China	Jul. 2024
GPU/server maintenance and junior-student mentoring	2025 – Present

TECHNICAL SKILLS

Programming: Python, C++, SQL, MATLAB; working knowledge of Java and JavaScript.
Frameworks and tools: PyTorch, NumPy, pandas, scikit-learn, TDengine, MySQL/SQLite, Git, Linux, LaTeX, pytest.
Methods: Counterfactual inference, off-policy evaluation, graph learning, time-series modeling, sequential detection, Taylor-model reachability, and quantitative backtesting.